

Artificial Intelligence in the Knowledge Economy

Ide & Talamàs (2025), Repository 5

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`github.com/wbgradost/ai-05-ide`

1. The question and the mechanism

Question. Who gains when AI agents enter a knowledge hierarchy, and does the answer depend on capability, autonomy, or both?

Production problems have unknown difficulty $x \sim U[0, 1]$. A human with knowledge z solves a fraction z alone. Workers handle routine problems; solvers handle exceptions that workers cannot solve.

Core mechanism

AI changes both the set of tasks that can be performed and the set of roles in which compute can be used. This changes matches, occupational composition, and the division of surplus.

2. The pre-AI economy

There is a unit mass of humans, one unit of time, and knowledge $z \in [0, 1]$.

- ▶ A single-layer producer works alone and expected output is z .
- ▶ A two-layer firm has workers of knowledge z and a solver of knowledge $s \geq z$.
- ▶ Each request for help costs the solver $h \in (0, 1)$ units of time.

$$n(z)h(1 - z) = 1, \quad n(z) = \frac{1}{h(1 - z)}.$$

Under the maintained case $h < h_0$, Proposition 1 gives the efficient, positively assortative hierarchy $W \preceq S$ with no independent human producers.

3. Capability and autonomy are different

An AI agent has fixed capability $z_{AI} < 1$ and uses one unit of compute.

	Autonomous AI	Non-autonomous AI
Independent production	yes	no
Worker/co-worker role	yes	no
Solver/co-pilot role	yes	yes
Compute price (abundant case)	$r = z_{AI}$	$r = 0$

Capability says how difficult a problem the agent can solve. Autonomy says whether it can pursue production without a human.

4. Autonomous equilibrium and occupational displacement

Autonomous AI is always used for independent production. It is also used as a worker when $z_{AI} \in \text{int}(W)$ and as a solver when $z_{AI} \in \text{int}(S)$ (Proposition 2, pp. 18–20).

- ▶ Basic AI: $z_{AI} \in \text{int}(W)$ displaces humans from routine work toward specialized problem solving.
- ▶ Advanced AI: $z_{AI} \in \text{int}(S)$ displaces humans in the opposite direction, toward routine work (Proposition 3, p. 21).

The same capability parameter therefore has different distributional effects depending on the occupation it enters.

5. Proposition 5: who wins under autonomous AI?

Let w be pre-AI wages and w^A autonomous-AI wages. Define

$$B = \{z \leq z_{AI} : w^A(z) > w(z)\}, \quad T = \{z \geq z_{AI} : w^A(z) > w(z)\}.$$

The appendix single-crossing lemma gives $B = [0, z_b)$ and $T = (z_t, 1]$. The agent exactly at z_{AI} loses because $w^A(z_{AI}) = z_{AI} < w(z_{AI})$.

Exact conditions (v11 p. 24; Appendix §2.5, pp. 18–21)

Bottom winners exist iff $z_{AI} > \bar{z}_{AI}$, where $\bar{z}_{AI} \in \text{int}(W)$. Top winners exist for every $z_{AI} \in [0, 1)$ under $h < h_0$.

6. Why the bottom threshold is real

For the least knowledgeable human, AI creates two forces:

$$\Delta(0) = w^A(0) - w(0) = \underbrace{\text{share effect}}_{+} - \underbrace{\text{match effect}}_{-} \quad (\text{basic AI}).$$

The proof constructs $\bar{z}_{AI}$ from the unique root $\Delta(0; \bar{z}_{AI}) = 0$. It shows $\Delta(0; z_{AI})$ is continuous and strictly increasing up to the worker/solver boundary. Hence

$$\Delta(0; z_{AI}) > 0 \quad \Longleftrightarrow \quad z_{AI} > \bar{z}_{AI}.$$

For advanced AI, both the match and share effects are positive, so the bottom wins automatically.

7. Why top winners are not capability-free

At the top, the share effect always dominates the match effect under the maintained assumptions. The proof uses

$$w^A(1) = \frac{1}{h} > w(1)$$

and the same single-crossing result.

This is not an unconditional statement about every model: the paper notes that top winners can disappear when $h \geq h_0$, and the superintelligent endpoint $z_{AI} = 1$ makes the most knowledgeable human lose.

Audit point: autonomy matters for the regime, but capability remains inside the winner conditions.

8. Proposition 6: non-autonomous AI

Let w^N denote non-autonomous-AI wages. AI can only be a solver, so $r = 0$ and unused compute is idle.

Adoption threshold (v11 p. 27; Appendix §2.6, pp. 21–27)

If $z_{AI} \leq w(0)$, AI is not used and occupations and wages equal the pre-AI equilibrium. If $z_{AI} > w(0)$, only the least knowledgeable humans use AI as a solver, through a unique cutoff allocation.

The threshold is explicitly about capability relative to the pre-AI bottom wage, not autonomy alone.

9. Proposition 6: distribution and output

Proposition 6 also establishes:

1. Autonomous-AI output is strictly higher than non-autonomous-AI output. Autonomous compute can produce; non-autonomous compute must leave some capacity idle.
2. With adoption, at least one human loses relative to pre-AI.
3. A neighborhood of the bottom does at least as well under non-autonomous AI as under pre-AI or autonomous AI (strictly better when adopted).
4. A neighborhood of the top does weakly worse under non-autonomous than under autonomous AI; the inequality is strict below the endpoint.

Bottom line: autonomy changes the opportunity-cost channel and output; capability controls adoption and the location of winners.

10. A two-type calculation to write by hand

Take two low types $z_L = 0$, one high type $z_H = 0.8$, and $h = 0.5$. Then $n(0) = 2$. Normalize the high type's outside option to $w_H = 0.8$.

$$Y^0 = 2(0.8) = 1.6, \quad w_L^0 = \frac{1.6 - 0.8}{2} = 0.4.$$

For capability $q \in (0.4, 0.8)$ and one compute unit:

$$Y^N = 0.8 + 2q, \quad Y^A = 1.6 + q,$$

because non-autonomous AI supervises the two low workers while autonomous AI uses compute independently and leaves the human hierarchy intact.

$$Y^A - Y^N = 0.8 - q > 0.$$

This is a transparent discrete specialization, not the continuum proof.

11. Where I did not believe the AI

The tempting summary was: “distributional effects depend on autonomy, not capability.”

Verdict

Too strong. Capability determines the autonomous bottom-winner threshold $z_{AI} > \bar{z}_{AI}$ and the non-autonomous adoption threshold $z_{AI} > w(0)$. Autonomy changes which roles compute can perform and whether compute has a production opportunity cost. The paper's two-dimensional taxonomy is more informative than the one-line gloss.

12. Handwritten derivation

Proposition 6

$$z_i = 0, z_n = \frac{4}{5}, h = \frac{1}{2}$$

$$n(0) = \frac{1}{h(1-z_i)} = \frac{1}{(\frac{1}{2})(1)} = 2$$

$$y^0 = 2 \cdot \left(\frac{4}{5}\right) = \frac{8}{5}$$

$$w_L^0 = \frac{\frac{8}{5} - \frac{4}{5}}{2} = \frac{2}{5}$$

$$\frac{2}{5} < q < \frac{4}{5}$$

$$y^N = \frac{4}{5} + 2q$$

$$y^A = \frac{8}{5} + q$$

$$y^A - y^N = \left(\frac{8}{5} + q\right) - \left(\frac{4}{5} + 2q\right) = \frac{4}{5} - q$$

$$q < \frac{4}{5} \rightarrow \frac{4}{5} - q > 0$$

$$\boxed{y^A > y^N}$$

13. Lean: original claim $\rightarrow$ formal statement

Original paper claim (Proposition 6, item 1):

$Y^A > Y^N$ when comparing autonomous and non-autonomous AI.

Our Lean target: a discrete specialization with two low types and one high type, for $2/5 < q < 4/5$,

$$\frac{4}{5} + 2q < \frac{8}{5} + q.$$

This is deliberately labelled a specialization: it is not the full continuum equilibrium statement.

14. Lean proof and interpretation

The transparent Spec is paired with theorem `discreteOutputStrictlyHigher`. It states $4/5 + 2*q < 8/5 + q$. The endpoint assumes $2/5 < q$ and $q < 4/5$, unfolds the definition, and closes the inequality with `linarith`.

What Lean verifies: given the explicit capability interval, the discrete autonomous-output expression exceeds the non-autonomous expression by $4/5 - q > 0$. The paper target compiled and the `AppliedModelingLib -fast` check passed.

What it does not verify: the paper's wage functions, continuum matching, Proposition 5's single-crossing threshold, or the full Proposition 6 equilibrium. The result is partial by design.

15. Takeaways

- ▶ Knowledge hierarchies turn AI adoption into a matching and surplus-sharing problem, not just a productivity shock.
- ▶ Autonomous AI always has a productive compute margin; non-autonomous AI may be unused and leaves compute idle.
- ▶ Proposition 5: top winners are guaranteed only under the maintained assumptions; bottom winners require $z_{AI} > \bar{z}_{AI}$.
- ▶ Proposition 6: non-autonomous adoption requires $z_{AI} > w(0)$ and shifts gains toward the bottom, while autonomous AI generates higher output.
- ▶ Lean checks one honest discrete output mechanism, complemented by the author's handwritten discrete derivation.